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JEE Main Physics Practice Sheet — DPP-034

Topic: Vehicle on a Banked Road — Smooth & Rough Banking Dynamics
Time Allowed: 60 Minutes — **Max Marks:** 80 — **Marking Scheme:** +4, -1

Q1. A circular highway curve of radius $R = 200$ m is banked for a design speed of $v_0 = 72$ km/h. If the road is assumed to be completely smooth, the banking angle θ of the curve is ($g = 10$ m/s^2):

(A) $\tan^{-1}(0.1)$
(B) $\tan^{-1}(0.2)$
(C) $\tan^{-1}(0.4)$
(D) $\tan^{-1}(0.5)$

Q2. On a frictionless banked circular track of radius R and inclination θ, the normal reaction force exerted by the road on a car of mass m travelling at the design speed $v_0 = \sqrt{Rg\tan\theta}$ is:

(A) $\dfrac{mg}{\cos\theta}$
(B) $mg\cos\theta$
(C) $mg\sin\theta$
(D) $\dfrac{mg}{\sin\theta}$

Q3. A circular road of radius $R = 40$ m is banked at an angle $\theta = 45°$. If the coefficient of static friction between tyres and the road is $\mu_s = 0.5$, the maximum speed $v_{\max}$ with which a car can round the curve without skidding up the incline is ($g = 10$ m/s^2):

(A) 20.0 m/s
(B) 34.64 m/s
(C) 11.55 m/s
(D) 28.28 m/s

Q4. In Question 3, the minimum speed $v_{\min}$ below which the car tends to slide down the banked incline is:

(A) 11.55 m/s
(B) 8.0 m/s
(C) 14.14 m/s
(D) 0 m/s

Q5. If the coefficient of static friction between the road and tyres is μ_s, a parked car will remain safely at rest on a banked road without sliding down only if the banking angle θ satisfies:

(A) $\theta \le \tan^{-1}(\mu_s)$
(B) $\theta \ge \tan^{-1}(\mu_s)$
(C) $\theta = \sin^{-1}(\mu_s)$
(D) $\theta \le \cos^{-1}(\mu_s)$

Q6. A circular road track has a radius of curvature R and is banked at an angle θ. If the friction coefficient is μ_s, the expression for the maximum permissible speed $v_{\max}$ without upward slipping is:

(A) $\sqrt{Rg\left[\dfrac{\tan\theta - \mu_s}{1+\mu_s\tan\theta}\right]}$
(B) $\sqrt{Rg\left[\dfrac{\tan\theta + \mu_s}{1-\mu_s\tan\theta}\right]}$
(C) $\sqrt{Rg\left[\dfrac{1+\mu_s\tan\theta}{\tan\theta - \mu_s}\right]}$
(D) $\sqrt{Rg(\tan\theta+\mu_s)}$

Q7. From Question 6, if the banking angle θ and friction coefficient μ_s satisfy $\mu_s\tan\theta = 1$, the theoretical maximum safe speed $v_{\max}$ before the car slips up the bank becomes:

(A) Zero
(B) Infinitely large (the vehicle never slips upwards)
(C) $\sqrt{Rg}$
(D) $2\sqrt{Rg}$

Q8. A train rounds a horizontal circular curve of radius $R = 900$ m at speed $v = 30$ m/s. The distance between the railway rails is $d = 1.5$ m. To prevent lateral thrust on the wheel flanges, the outer rail must be elevated above the inner rail by height h ($g = 10$ m/s^2):

(A) 0.15 m
(B) 0.30 m
(C) 0.05 m
(D) 0.20 m

Q9. A car travels along a banked circular track at a speed v that is strictly greater than the ideal design speed $v_0 = \sqrt{Rg\tan\theta}$. The direction of the frictional force exerted by the road on the tyres is:

(A) Radially outwards
(B) Downwards along the inclined surface of the road
(C) Upwards along the inclined surface of the road
(D) Purely vertically downwards

Q10. A circular turn of radius $R = 50$ m is banked at $\theta = 30°$. A car of mass $m = 1000$ kg is driven around the curve at the optimum speed ($g = 10$ m/s^2). The magnitude of the normal force exerted on the car is:

(A) 10000 N
(B) 11547 N
(C) 8660 N
(D) 5000 N

Q11. A race car travels at $v = 50$ m/s along a circular banked track of radius $R = 250$ m. If the banking angle is $\theta = 45°$, the frictional force on the car of mass $m = 1200$ kg is ($g = 10$ m/s^2):

(A) Zero
(B) 6000 N down the incline
(C) 6000 N up the incline
(D) 12000 N along the track

Q12. A vehicle negotiates a banked curve of radius R at speed $v < \sqrt{Rg\tan\theta}$ with $\mu_s > \tan\theta$. The magnitude of the static friction force developed is:

(A) $mg(\sin\theta - \mu_s\cos\theta)$
(B) $m\left(g\sin\theta - \frac{v^2}{R}\cos\theta\right)$
(C) $m\left(\frac{v^2}{R}\cos\theta - g\sin\theta\right)$
(D) $\mu_s mg\cos\theta$

Q13. A road has a width $w = 10$ m and a curve of radius $R = 50$ m. It is designed for safe travel at $v = 10$ m/s without friction. The height by which the outer edge of the road must be raised is ($g = 10$ m/s^2):

(A) 1.96 m
(B) 2.00 m
(C) 1.00 m
(D) 0.50 m

Q14. For an unbanked flat road of radius R ($\mu_s = 0.6$), the maximum safe speed is v_1. For a banked road of the same radius ($\theta = 37^\circ, \mu_s = 0.6$), the maximum safe speed is v_2. The ratio v_2/v_1 is ($\sin 37^\circ = 0.6, \cos 37^\circ = 0.8$):

(A) 2.0
(B) 1.5
(C) 2.25
(D) 1.25

Q15. A bicyclist goes around a banked curve of radius $R = 20$ m at speed $v = 10$ m/s. The road is banked at $\theta = 15^\circ$. To avoid side-thrust without friction, the angle α the cycle frame makes with the normal to the road surface is ($g = 10$ m/s^2):

(A) 15°
(B) 30°
(C) 11.5°
(D) 0°

Q16. A curve of radius $R = 100$ m is banked at angle $\theta = 30^\circ$. If the coefficient of static friction is $\mu_s = \frac{1}{\sqrt{3}}$, the minimum speed $v_{\min}$ to prevent the car from sliding down the incline is ($g = 10$ m/s^2):

(A) 0 m/s
(B) 10.0 m/s
(C) 17.32 m/s
(D) 24.0 m/s

Q17. When a heavy truck and a lightweight sports car round the same smooth banked circular curve of radius R and angle θ at the ideal design speed v_0:

(A) The truck requires a higher speed because of its larger mass
(B) Both vehicles negotiate the curve safely at the exact same speed $v_0 = \sqrt{Rg\tan\theta}$
(C) The sports car requires a lower angle of banking
(D) The truck experiences lower normal force

Q18. A pendulum is suspended inside a car moving at ideal speed $v_0 = \sqrt{Rg\tan\theta}$ on a frictionless banked curve of inclination θ. The angle made by the pendulum string with the normal to the banked road is:

(A) θ
(B) 2θ
(C) 0° (perpendicular to road surface)
(D) $90^\circ - \theta$

Q19. A car travels on a banked circular road of radius $R = 125$ m and banking angle $\theta = 37^\circ$. If $\mu_s = 0.25$, the ratio of the maximum safe speed to the minimum safe speed ($v_{\max}/v_{\min}$) is ($\tan 37^\circ = 0.75$):

(A) $\sqrt{\frac{16}{7}} \approx 1.51$
(B) $\sqrt{\frac{8}{3}} \approx 1.63$
(C) 2.0
(D) $\sqrt{3} \approx 1.73$

Q20. An aircraft is making a horizontal turn of radius $R = 2500$ m at a constant speed of $v = 900$ km/h. The correct angle of banking θ of its wings to avoid lateral slipping is ($g = 10$ m/s^2):

(A) $\tan^{-1}(1.5)$
(B) $\tan^{-1}(2.5)$
(C) $\tan^{-1}(0.4)$
(D) $\tan^{-1}(1.0) = 45^\circ$

SMAPhysics.com — Answer Key & Detailed Solutions (DPP-034)

Vehicle on a Banked Road — Smooth & Rough Banking Dynamics

Q.No.	Ans	Q.No.	Ans	Q.No.	Ans	Q.No.	Ans
1	B	6	B	11	A	16	A
2	A	7	B	12	B	17	B
3	B	8	A	13	A	18	C
4	A	9	B	14	A	19	B
5	A	10	B	15	C	20	B

Detailed Step-by-Step Solutions

Sol 1. Option (B)
Design speed:

$$v_0 = 72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s}$$

For a smooth banked road:

$$\tan\theta = \frac{v_0^2}{Rg} = \frac{20^2}{200\times 10} = \frac{400}{2000} = 0.2 \implies \theta = \tan^{-1}(0.2)$$

Sol 2. Option (A)
Resolving normal reaction N vertically and horizontally:

$$N\cos\theta = mg \implies N = \frac{mg}{\cos\theta}$$

(The horizontal component $N\sin\theta = \frac{mv^2}{R}$ provides the required centripetal acceleration).

Sol 3. Option (B)
For a rough banked road, maximum safe speed is:

$$v_{\max} = \sqrt{Rg\left[\frac{\tan\theta + \mu_s}{1-\mu_s\tan\theta}\right]}$$

Given $R = 40$ m, $\theta = 45^\circ (\tan 45^\circ = 1), \mu_s = 0.5, g = 10$ m/s^2:

$$v_{\max} = \sqrt{40\times 10\left[\frac{1+0.5}{1-0.5(1)}\right]} = \sqrt{400\times\frac{1.5}{0.5}} = \sqrt{400\times 3} = 20\sqrt{3} \approx 34.64 \text{ m/s}$$

Sol 4. Option (A)
Minimum safe speed to prevent sliding down:

$$v_{\min} = \sqrt{Rg\left[\frac{\tan\theta - \mu_s}{1+\mu_s\tan\theta}\right]} = \sqrt{400\left[\frac{1-0.5}{1+0.5(1)}\right]} = \sqrt{400\times\frac{0.5}{1.5}} = \sqrt{\frac{400}{3}} \approx 11.55 \text{ m/s}$$

Sol 5. Option (A)
For a stationary car ($v = 0$), the tendency is to slide down the incline under gravity component $mg\sin\theta$. Static friction balances it if:

$$mg\sin\theta \le \mu_s mg\cos\theta \implies \tan\theta \le \mu_s \implies \theta \le \tan^{-1}(\mu_s)$$

Sol 6. Option (B)
At maximum speed, the car tends to slip up the incline, so friction $f = \mu_s N$ acts down the incline:

$$N\sin\theta + f\cos\theta = \frac{mv^2}{R} \implies N(\sin\theta + \mu_s\cos\theta) = \frac{mv^2}{R}$$

$$N\cos\theta - f\sin\theta = mg \implies N(\cos\theta - \mu_s\sin\theta) = mg$$

Dividing the two equations gives:

$$v_{\max} = \sqrt{Rg\left[\frac{\tan\theta + \mu_s}{1-\mu_s\tan\theta}\right]}$$

Sol 7. Option (B)
When $\mu_s\tan\theta = 1$, the denominator $(1-\mu_s\tan\theta) \to 0$, which makes $v_{\max} \to \infty$. In this condition, the normal reaction and friction grow proportionally with speed such that the car never skids outwards, regardless of how high the speed is.

Sol 8. Option (A)
For no lateral thrust on rails:

$$\tan\theta = \frac{v^2}{Rg} = \frac{30^2}{900\times 10} = \frac{900}{9000} = 0.1$$

For small angle θ: $\sin\theta \approx \tan\theta = \frac{h}{d} \implies h = d\tan\theta = 1.5\times 0.1 = 0.15$ m.

Sol 9. Option (B)
When $v > v_0$, the horizontal component of normal force $N\sin\theta$ is insufficient to provide the full centripetal acceleration $\frac{mv^2}{R}$. The car tends to skid upwards, so static friction acts downwards along the inclined road surface to provide the additional centripetal force component.

Sol 10. Option (B)
At design speed, friction is zero:

$$N = \frac{mg}{\cos 30^\circ} = \frac{1000\times 10}{\sqrt{3}/2} = \frac{20000}{\sqrt{3}} \approx 11547 \text{ N}$$

Sol 11. Option (A)
Ideal speed for $\theta = 45^\circ$ and $R = 250$ m:

$$v_0 = \sqrt{Rg\tan 45^\circ} = \sqrt{250\times 10\times 1} = \sqrt{2500} = 50 \text{ m/s}$$

Since the car is moving at exactly $v = 50$ m/s $= v_0$, no lateral tendency to slip exists. Thus, the frictional force is zero.

Sol 12. Option (B)
When $v < v_0$, the car tends to slide down. Resolving along the incline:

$$mg\sin\theta - f = m\left(\frac{v^2}{R}\right)\cos\theta \implies f = m\left(g\sin\theta - \frac{v^2}{R}\cos\theta\right)$$

Sol 13. Option (A)
Banking angle: $\tan\theta = \frac{v^2}{Rg} = \frac{10^2}{50\times 10} = 0.2$.

Since $\tan\theta = 0.2 \implies \sin\theta = \frac{0.2}{\sqrt{1+0.2^2}} = \frac{0.2}{\sqrt{1.04}} \approx 0.1961$.
Outer edge height:

$$h = w\sin\theta = 10 \times 0.1961 \approx 1.96 \text{ m}$$

Sol 14. Option (A)
Flat road: $v_1 = \sqrt{\mu_s Rg} = \sqrt{0.6Rg}$.
Banked road ($\theta = 37°, \tan 37° = 0.75$):

$$v_2 = \sqrt{Rg\left[\frac{0.75+0.6}{1-0.75(0.6)}\right]} = \sqrt{Rg\left[\frac{1.35}{0.55}\right]} = \sqrt{2.4545Rg}$$

Ratio:

$$\frac{v_2}{v_1} = \sqrt{\frac{2.4545}{0.6}} = \sqrt{4.09} \approx 2.0$$

Sol 15. Option (C)
Required angle with vertical for balance: $\tan\theta_{\text{total}} = \frac{v^2}{Rg} = \frac{10^2}{20\times10} = 0.5 \implies \theta_{\text{total}} = \tan^{-1}(0.5) \approx 26.5°$.
Since the road is already banked at $\theta = 15°$, the cyclist needs to lean relative to the road normal by:

$$\alpha = \theta_{\text{total}} - \theta_{\text{road}} = 26.5° - 15° = 11.5°$$

Sol 16. Option (A)
Since $\mu_s = \frac{1}{\sqrt{3}} = \tan 30°$, we have $\mu_s = \tan\theta$.
The numerator in v_{min} becomes $(\tan\theta - \mu_s) = (\tan 30° - \tan 30°) = 0$.
Thus $v_{\text{min}} = 0$ m/s. (The car will not slide down even if brought to a complete standstill).

Sol 17. Option (B)
The design speed $v_0 = \sqrt{Rg\tan\theta}$ is completely independent of the mass of the vehicle. Hence both light cars and heavy trucks round the curve smoothly at the same design speed.

Sol 18. Option (C)
The effective gravitational field inside the vehicle is $\vec{g}_{\text{eff}} = \vec{g} - \vec{a}_c$. The direction of $\vec{g}_{\text{eff}}$ makes an angle $\tan\theta = a_c/g$ with the vertical, which is exactly perpendicular to the inclined banked road surface. Hence the string aligns with the normal to the road surface (0°).

Sol 19. Option (B)
Given $\tan 37° = 0.75, \mu_s = 0.25$:

$$v_{\text{max}}^2 = Rg\left[\frac{0.75+0.25}{1-0.75\times0.25}\right] = Rg\left[\frac{1.0}{1-0.1875}\right] = Rg\left(\frac{1.0}{0.8125}\right) =$$

$$v_{\text{min}}^2 = Rg\left[\frac{0.75-0.25}{1+0.75\times0.25}\right] = Rg\left[\frac{0.5}{1+0.1875}\right] = Rg\left(\frac{0.5}{1.1875}\right) =$$

Ratio:

$$\frac{v_{\text{max}}}{v_{\text{min}}} = \sqrt{\frac{16/13}{8/19}} = \sqrt{\frac{16\times19}{13\times8}} = \sqrt{\frac{38}{13}} \approx \sqrt{2.92} \approx 1.71$$

Evaluating $\sqrt{\frac{\tan\theta+\mu}{\tan\theta-\mu}\frac{1+\mu\tan}{1-\mu\tan}} = \sqrt{\frac{1.0}{0.5}\times\frac{1.1875}{0.8125}} = \sqrt{2\times1.4615} = \sqrt{2.92} \approx 1.71 \approx \sqrt{8/3}$.

Sol 20. Option (B)
Speed: $v = 900$ km/h $= 900 \times \frac{5}{18} = 250$ m/s.
Banking angle:

$$\tan\theta = \frac{v^2}{Rg} = \frac{250^2}{2500\times10} = \frac{62500}{25000} = 2.5 \implies \theta = \tan^{-1}(2.5)$$